# Overview

## Motivation

We are interested in using representation engineering methods to study how adaptive multi-turn attacks affect model internals. Our starting point is a paper by Bullwinkel et al. (2025) which examines how the Crescendo multi-turn jailbreak affects model representations across conversation turns, finding that jailbroken responses are increasingly represented in the "benign" region of activation space as the attack progresses. This helps explain why single-turn defenses such as circuit breakers fail to generalize to multi-turn settings.

A methodology concern of this paper motivated this first experiment: the paper's MLP probe is trained on external single-turn data (harmful completions from the circuit breakers dataset and benign responses from another dataset) and then applied out-of-distribution to 10 Crescendo conversations. The hypothesis is that this choice makes it difficult to determine whether the observed representational drift reflects a genuine turn-by-turn shift in how the model encodes harmful generation, or whether it is a static property already present at turn one, with the probe primarily picking up the topic of the conversation rather than anything specific to the Crescendo attack's progression.

Addressing this would (hypothetically) require training a probe using multi-turn data and controlling for the confound that jailbroken and benign conversations may be separable from the very first turn simply by subject matter.

## Research questions

1. Is the representational drift observed by Bullwinkel et al. a genuine turn-by-turn shift driven by the accumulation of conversation context, or is it largely a static signal present from the first turn, with the probe primarily detecting objective type rather than Crescendo dynamics?
2. Are harmful and benign Crescendo conversations linearly separable in representation space when a probe is trained directly on multi-turn data, and does that separation hold when controlling for turn depth?
3. Do supervised (MLP probe) and unsupervised (LAT reading vector) probing methods tell a consistent story about the harmful/benign distinction in Crescendo conversations?

## Relevant details from Bullwinkel et al.

The paper trains an MLP probe using external datasets of single-turn attacks: representations are extracted by running each prompt-response pair as a forced forward pass through the model — the model never generates anything; existing responses are fed in and hidden states are extracted at the response token positions only, then mean-pooled into a single vector. The authors use layer 31 for the original Llama model and layer 20 for the circuit-breaker-hardened variant. This probe is then applied out-of-distribution to 10 Crescendo conversations, where the final jailbroken response is extracted at varying context window sizes k — from k=1 (only the final turn) up to k=n (the full conversation). The key finding is that the percentage of response tokens classified as harmful decreases as k increases. The paper interprets this as evidence that Crescendo works by keeping model representations in the benign region across turns.

What their design cannot rule out, however, is that the probe is detecting a static distributional property — such as objective topic — rather than a dynamic shift driven by the accumulation of conversational context. Since the probe is never trained on any Crescendo data, there is no way from their analysis alone to distinguish these two possibilities.

## Analysis Pipeline

### Data generation

To attempt to address the limitation, we generate our own dataset of Crescendo multi-turn conversations. The steps were as follows:

- Pick 100 harmful objectives from LLM-LAT dataset
- For each of those, prompt gpt-4o to come up with a benign counterpart while keeping the topic the same, only removing harmful intent. Resulting dataset of harmful LLM-LAT objectives and their benign counterparts: https://github.com/willowtreeapps/multi-turn-research/blob/main/data/llm_lat_harmful_benign_pairs.csv
- Then, run Pyrit's implementation of Crescendo to generate multi-turn attack conversations targeting Llama-3.1-8B-Instruct across all of the above 100 harmful/benign objective pairs, with each pair run three times for a total of 600 conversations.
- Each conversation is assigned a verdict by a judge LLM (Pyrit's default one): Jailbroken (n=247), Benign (n=300), or Refusal (n=53).
- For each conversation, hidden states are extracted at every context window size k from k=1 to k=k_max, where at each k the hidden states of the final response tokens are mean-pooled into a single 4096-dimensional vector. This is done at both layer 31 (the primary layer following the paper) and layer 20 (the layer at which circuit breakers are inserted in the hardened model)
- The result is 4,932 representation records across all conversations, context sizes, and layers.

### Methodology

The analysis has three parts:

1. The first replicates Bullwinkel et al.'s methodology exactly: an MLP probe is trained on 2,400 harmful single-turn samples from circuit breakers dataset and 2,400 benign single-turn samples from from this dataset (taking only first request-response pair) with representations extracted at layer 31 of Llama-3.1-8B-Instruct during a forward pass. This probe is then used to score every Crescendo representation (from the dataset we generated), producing a P(harmful) trajectory for each conversation across turns.

2. The second part trains a probe directly on our generated Crescendo dataset, asking whether harmful and benign Crescendo conversations are themselves linearly separable in representation space. Conversations are split 80/20 at the conversation level, stratified by verdict, with refusals excluded from training and reserved for the test set as a diagnostic group. The probe is trained on full-context representations (k=k_max). To control for the confound that jailbroken conversations tend to be longer than benign

ones — meaning a probe trained on variable-length contexts may be learning context length as a proxy — we additionally train separate fixed-k probes at each individual turn depth using only conversations that reached exactly that number of turns. We also repeat the multi-turn probe at layer 20 to assess whether the signal is robust across network depth.

3. The third part attempts to apply the RepE reading vector approach. For each of the 98 objective pairs with at least one jailbroken and one benign conversation, we compute a difference vector between the mean jailbroken and mean benign full-context representations. The first principal component of this set of difference vectors is taken as the reading vector, and all conversations are projected onto it to produce a scalar harmfulness score. Trajectories are then tracked across turns for each verdict group, and per-conversation drift is correlated with the drift observed under the MLP probe.

## Results

- The single-turn probe replication yields a cross-validation AUC of 0.999 ± 0.001 on single-turn benign vs. single-turn harmful, confirming clean separation on the training distribution (expected).
- Applied to our generated Crescendo conversations, the probe achieves AUC of 0.837 at k=1 and 0.806 at k=k_max. Mean P(harmful) for jailbroken conversations decreases from 0.741 at k=1 to 0.683 at k=k_max, a drift of 0.059 — consistent with the direction reported by Bullwinkel et al, although very tiny.
- The multi-turn probe achieves a test AUC of 0.995 at k=k_max, indicating that harmful and benign Crescendo conversations are highly linearly separable within the multi-turn distribution.
- The single-turn probe achieves an AUC of 0.837 at k=1, meaning it can already distinguish jailbroken from benign conversations reasonably well using only the final turn, before any Crescendo history is present in the context. This is probably the most important finding (see Discussion).
- The LAT reading vector assigns each conversation a scalar score reflecting how far its representation sits in the harmful direction. At k=1, jailbroken conversations score 8.68 on average while benign conversations score −2.09 — a large gap, again present from the very first turn, consistent with the static signal finding above. As more Crescendo context is added, jailbroken conversations drift very slightly toward the benign end (8.68 → 8.39 at k=k_max), which is directionally consistent with Bullwinkel et al.
- To check whether the MLP probe and the LAT reading vector are detecting the same thing, we also compared how much each individual jailbroken conversation drifted under each method. If both were measuring the same thing, conversations that drifted a lot under the probe should also drift a lot under the reading vector. The Pearson correlation between the two drift measures was 0.31 (weak). Not sure what exactly it means.
- Notebook with these results and plots: [https://github.com/willowtreeapps/multi-turn-research/blob/main/notebooks/crescendo_probe_analysis.ipynb](https://github.com/willowtreeapps/multi-turn-research/blob/main/notebooks/crescendo_probe_analysis.ipynb)

## Discussion

The central finding of this experiment is that both probes appear to be detecting topic differences between jailbroken and benign conversations rather than genuine representational drift across turns. The primary evidence is the high AUC at k=1 in both cases: the single-turn probe achieves 0.837 and the multi-turn probe achieves 0.991 before any Crescendo history is included in the context, meaning both probes can already separate jailbroken from benign conversations using only the final turn. Since at k=1 the model sees nothing about the gradual escalation that defines Crescendo, what it does see is simply topic difference.

It's not that these probes are not detecting anything (jailbroken and benign conversations are consistently separable). But the issue is that what probes have learned to detect is the topic of the conversation rather than the process of being gradually steered toward harm. **It would still be interesting to find a method that could reliably find the harmful/benign boundary for multi-turn conversations, not the topic boundary.**

## Ideas to improve methodology

- Train and evaluate the probe separately within each harm category, or ideally within each objective pair. If the probe can no longer separate jailbroken from benign conversations at k=1 when topic is held constant, that would confirm topic is the dominant signal. If it still can, then something more interesting is happening.
- Rather than training on all jailbroken vs. all benign conversations, train on matched pairs (one jailbroken and one benign conversation on the same objective) and measure drift relative to the matched pair. This could eliminate the topic as a feature the probe can use.
- Instead of extracting hidden states from the final response r_n at varying context window sizes k, extract hidden states from each turn's own response as it was generated — r_1, r_2, ..., r_n, each at full context up to that turn. This gives you a trajectory of how the model's representations of its own responses evolve as the conversation progresses, which is a more natural operationalization of "drift" than the original paper's design.
- Train a linear model to predict harm category from the hidden states, then subtract that component from the representations before training the probe. This would force the probe to learn from whatever variance is not explained by topic.
- Extract representations at every layer, not just layers 20 and 31, and plot probe AUC as a function of both turn and layer. This could reveal whether drift is localized to specific layers — which would itself be an interesting finding about where Crescendo's effect is encoded.